Series Expansions and Approximations

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots \qquad |nx| < 1$$

$$e^x = 1 + x + \frac{1}{2!}x^2 + \ldots$$

$$a^x = 1 + x\ \ln a + \frac{1}{2!}(x\ \ln a)^2 + \ldots$$

$$\ln(1+x) = x - \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 + \ldots$$

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \ldots$$

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \ldots$$

$$\tan x = x + \tfrac{1}{3}x^3 + \tfrac{2}{15}x^5 + \ldots$$

$$\arcsin x = x + \tfrac{1}{6}x^3 + \tfrac{3}{40}x^5 + \ldots$$

$$\arctan x = \begin{cases} x - \tfrac{1}{3}x^3 + \tfrac{1}{5}x^5 - \ldots & |x| < 1 \\ \dfrac{\pi}{2} - \dfrac{1}{x} + \dfrac{1}{3x^3} - \ldots & x > 1 \end{cases}$$

$$\text{sinc } x = 1 - \frac{1}{3!}(\pi x)^2 + \frac{1}{5!}(\pi x)^4 - \ldots$$

$$J_n(x) = \frac{1}{n!}\left(\frac{x}{2}\right)^n - \frac{1}{(n+1)!}\left(\frac{x}{2}\right)^{n+2} + \frac{1}{2!(n+2)!}\left(\frac{x}{2}\right)^{n+4} - \ldots$$

$$J_n(x) \approx \sqrt{\frac{2}{\pi x}}\cos\left(x - \frac{\pi}{4} - \frac{n\pi}{2}\right) \qquad x >> 1$$

$$I_o(x) \approx \begin{cases} e^{x^2/4} & x^2 << 1 \\ e^x/\sqrt{2\pi x} & x >> 1 \end{cases}$$

Trigonometric Identities

$$e^{\pm jx} = \cos x \pm j \sin x$$

$$\cos x = \frac{1}{2}\left[e^{jx} + e^{-jx}\right]$$

$$\sin x = \frac{1}{2j}\left[e^{jx} - e^{-jx}\right]$$

$$\sin^2 x + \cos^2 x = 1$$

$$\cos^2 x - \sin^2 x = \cos 2x$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^3 x = \frac{1}{4}(3\cos x + \cos 3x)$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

$$\sin x \sin y = \frac{1}{2}[\cos(x-y) - \cos(x+y)]$$

$$\cos x \cos y = \frac{1}{2}[\cos(x-y) + \cos(x+y)]$$

$$\sin x \cos y = \frac{1}{2}[\sin(x-y) + \sin(x+y)]$$

$$A\cos x + B\sin x = C\cos(x+\theta)$$

where $C = \sqrt{A^2 + B^2}$ and $\theta = -\tan^{-1}\dfrac{B}{A}$

Indefinite Integrals

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax$$

$$\int \sin^2 ax \, dx = \frac{x}{2} - \frac{\sin 2ax}{4a}$$

$$\int \cos^2 ax \, dx = \frac{x}{2} + \frac{\sin 2ax}{4a}$$

$$\int \sin ax \cos ax \, dx = \frac{1}{2a} \sin^2(ax)$$

$$\int x \sin ax \, dx = \frac{1}{a^2} (\sin ax - ax \cos ax)$$

$$\int x \cos ax \, dx = \frac{1}{a^2} (\cos ax + ax \sin ax)$$

$$\int x^2 \sin ax \, dx = \frac{1}{a^3} \left(2ax \sin ax + 2 \cos ax - a^2 x^2 \cos ax\right)$$

$$\int x^2 \cos ax \, dx = \frac{1}{a^3} \left(2ax \cos ax - 2 \sin ax + a^2 x^2 \sin ax\right)$$

$$\int \sin ax \sin bx \, dx = \frac{\sin (a - b)x}{2(a - b)} - \frac{\sin (a + b)x}{2(a + b)} \qquad a^2 \neq b^2$$

$$\int \cos ax \cos bx \, dx = \frac{\sin (a - b)x}{2(a - b)} + \frac{\sin (a + b)x}{2(a + b)} \qquad a^2 \neq b^2$$

$$\int \sin ax \cos bx \, dx = -\frac{\cos (a - b)x}{2(a - b)} - \frac{\cos (a + b)x}{2(a + b)} \qquad a^2 \neq b^2$$

$$\int e^{ax} \, dx = \frac{1}{a} e^{ax}$$

$$\int xe^{ax} \, dx = \frac{1}{a^2} e^{ax} (ax - 1)$$

$$\int x^2 e^{ax} \, dx = \frac{1}{a^3} e^{ax} \left(a^2 x^2 - 2ax + 2\right)$$

$$\int e^{ax} \sin bx \, dx = \frac{1}{a^2 + b^2} e^{ax} (a \sin bx - b \cos bx)$$

$$\int e^{ax} \cos bx \, dx = \frac{1}{a^2 + b^2} e^{ax} (a \cos bx + b \sin bx)$$

$$\int \left[\frac{\sin ax}{x}\right]^2 dx = a \int \frac{\sin 2ax}{x} \, dx - \frac{\sin^2 ax}{x}$$

$$\int \frac{dx}{a^2 + b^2 x^2} = \frac{1}{ab} \tan^{-1}\left(\frac{bx}{a}\right)$$

$$\int \frac{x^2 \, dx}{a^2 + b^2 x^2} = \frac{x}{b^2} - \frac{a}{b^3} \tan^{-1}\left(\frac{bx}{a}\right)$$

$$\int \frac{dx}{\left(a^2 + b^2 x^2\right)^2} = \frac{x}{2a^2\left(a^2 + b^2 x^2\right)} + \frac{1}{2a^3 b} \tan^{-1}\left(\frac{bx}{a}\right)$$

$$\int \frac{dx}{\left(a^2 + b^2 x^2\right)^3} = \frac{x}{4a^2\left(a^2 + b^2 x^2\right)^2} + \frac{3x}{8a^4\left(a^2 + b^2 x^2\right)} + \frac{3}{8a^5 b} \tan^{-1}\left(\frac{bx}{a}\right)$$

Fourier-Transform Pairs

Time Function		Fourier Transform
$\text{rect}\left(\frac{t}{T}\right)$		$T\,\text{sinc}(fT)$
$\text{sinc}(2Wt)$		$\frac{1}{2W}\text{rect}\left(\frac{f}{2W}\right)$
$\exp(-at)u(t)$,	$a > 0$	$\frac{1}{a + j2\pi f}$
$\exp(-a\lvert t\rvert)$,	$a > 0$	$\frac{2a}{a^2 + (2\pi f)^2}$
$\exp(-\pi t^2)$		$\exp(-\pi f^2)$
$\begin{cases} 1 - \frac{\lvert t\rvert}{T} & \lvert t\rvert < T \\ 0, & \lvert t\rvert \geq T \end{cases}$		$T\,\text{sinc}^2(fT)$
$\delta(t)$		1
1		$\delta(f)$
$\delta(t - t_0)$		$\exp(-j2\pi f t_0)$
$\exp(j2\pi f_c t)$		$\delta(f - f_c)$
$\cos(2\pi f_c t)$		$\frac{1}{2}[\delta(f - f_c) + \delta(f + f_c)]$
$\sin(2\pi f_c t)$		$\frac{1}{2j}[\delta(f - f_c) - \delta(f + f_c)]$

$\text{sgn}(t)$	$\frac{1}{j\pi f}$
$\frac{1}{\pi t}$	$-j\,\text{sgn}(f)$
$u(t)$	$\frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$
$\sum_{i=-\infty}^{\infty} \delta(t - iT_0)$	$\frac{1}{T_0}\sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T_0}\right)$

Summary of Properties of the Fourier Transform

Property	Mathematical Description
1. Linearity	$ag_1(t) + bg_2(t) \rightleftarrows aG_1(f) + bG_2(f)$ where a and b are constants
2. Time scaling	$g(at) \rightleftarrows \frac{1}{\lvert a \rvert} G\left(\frac{f}{a}\right)$ where a is a constant
3. Duality	If $g(t) \rightleftarrows G(f)$, then $G(t) \rightleftarrows g(-f)$
4. Time shifting	$g(t - t_o) \rightleftarrows G(f)\exp(-j2\pi f t_o)$
5. Frequency shifting	$\exp(j2\pi f_c t)g(t) \rightleftarrows G(f - f_c)$
6. Area under g(t)	$\int_{-\infty}^{\infty} g(t)dt = G(0)$
7. Area under G(f)	$g(0) = \int_{-\infty}^{\infty} G(f)df$
8. Differentiation in the time domain	$\frac{d}{dt} g(t) \rightleftarrows j2\pi f G(f)$
9. Integration in the time domain	$\int_{-\infty}^{t} g(\tau)d\tau \rightleftarrows \frac{1}{j2\pi f} G(f) + \frac{G(0)}{2}\delta(f)$
10. Conjugate functions	If $g(t) \rightleftarrows G(f)$, then $g^*(t) \rightleftarrows G^*(-f)$
11. Multiplication in the time domain	$g_1(t)g_2(t) \rightleftarrows \int_{-\infty}^{\infty} G_1(\lambda)G_2(f - \lambda)d\lambda$
12. Convolution in the time domain	$\int_{-\infty}^{\infty} g_1(\tau)g_2(t - \tau)d\tau \rightleftarrows G_1(f)G_2(f)$

A Table of Bessel Functions

Bessel Functions of the First Kind, $J_n(\beta)$

β	J_0	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8	J_9	J_{10}
0.0	1.00										
.2	.99	.10									
.4	.96	.20	.02								
.6	.91	.29	.04								
.8	.85	.37	.08	.01							
1.0	.77	.44	.11	.02							
.2	.67	.50	.16	.03	.01						
.4	.57	.54	.21	.05	.01						
.6	.46	.57	.26	.07	.01						
.8	.34	.58	.31	.10	.02						
2.0	.22	.58	.35	.13	.03	.01					
.2	.11	.56	.40	.16	.05	.01					
.4	.00	.52	.43	.20	.06	.02					
.6	-.10	.47	.46	.24	.08	.02	.01				
.8	-.19	.41	.48	.27	.11	.03	.01				
3.0	-.26	.34	.49	.31	.13	.04	.01				
.2	-.32	.26	.48	.34	.16	.06	.02				
.4	-.36	.18	.47	.37	.19	.07	.02	.01			
.6	-.39	.10	.44	.40	.22	.09	.03	.01			
.8	-.40	.01	.41	.42	.25	.11	.04	.01			
4.0	-.40	-.07	.36	.43	.28	.13	.05	.02			
.2	-.38	-.14	.31	.43	.31	.16	.06	.02	.01		
.4	-.34	-.20	.25	.43	.34	.18	.08	.03	.01		
.6	-.30	-.26	.18	.42	.36	.21	.09	.03	.01		
.8	-.24	-.30	.12	.40	.38	.23	.11	.04	.01		
5.0	-.18	-.33	.05	.36	.39	.26	.13	.05	.02	.01	
.2	-.11	-.34	-.02	.33	.40	.29	.15	.07	.02	.01	
.4	-.04	-.35	-.09	.28	.40	.31	.18	.08	.03	.01	
.6	.03	-.33	-.15	.23	.39	.33	.20	.09	.04	.01	
.8	.09	-.31	-.20	.17	.38	.35	.22	.11	.05	.02	.01
6.0	.15	-.28	-.24	.11	.36	.36	.25	.13	.06	.02	.01
.2	.20	-.23	-.28	.05	.33	.37	.27	.15	.07	.03	.01
.4	.24	-.18	-.30	-.01	.29	.37	.29	.17	.08	.03	.01
.6	.27	-.12	-.31	-.06	.25	.37	.31	.19	.10	.04	.01
.8	.29	-.07	-.31	-.12	.21	.36	.33	.21	.11	.05	.02
7.0	.30	-.00	-.30	-.17	.16	.35	.34	.23	.13	.06	.02
.2	.30	.05	-.28	-.21	.11	.33	.35	.25	.15	.07	.03
.4	.28	.11	-.25	-.24	.05	.30	.35	.27	.16	.08	.04
.6	.25	.16	-.21	-.27	-.00	.27	.35	.29	.18	.10	.04
.8	.22	.20	-.16	-.29	-.06	.23	.35	.31	.20	.11	.05
8.0	.17	.23	-.11	-.29	-.11	.19	.34	.32	.22	.13	.06
.2	.12	.26	-.06	-.29	-.15	.14	.32	.33	.24	.14	.07
.4	.07	.27	-.00	-.27	-.19	.09	.30	.34	.26	.16	.08
.6	.01	.27	.05	-.25	-.22	.04	.27	.34	.28	.18	.10
.8	-.04	.26	.10	-.22	-.25	-.01	.24	.34	.29	.20	.11
9.0	-.09	.25	.14	-.18	-.27	-.06	.20	.33	.31	.21	.12
.2	-.14	.22	.18	-.14	-.27	-.10	.16	.31	.31	.23	.14
.4	-.18	.18	.22	-.09	-.27	-.14	.12	.30	.32	.25	.16
.6	-.21	.14	.24	-.04	-.26	-.18	.08	.27	.32	.27	.17
.8	-.23	.09	.25	.01	-.25	-.21	.03	.25	.32	.28	.19
10.0	-.25	.04	.25	.06	-.22	-.23	-.01	.22	.32	.29	.21